

NIPUN ARORA

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SUMMARY

Computer Science (AI/ML) undergraduate with strong focus on building real-world full-stack systems and deploying them in production environments. Experienced in backend development, cloud storage (AWS S3), containerization (Docker), and system design. Focused on building end-to-end systems involving APIs, infrastructure, and user-facing applications.

EDUCATION

Bachelor of Technology (Computer Science Engineering - AI/ML)

Bennett University · 2023 – 2027 · CGPA - 8.22

Research

Benchmarking Explainability for Pneumonia Detection in Chest X-Rays

Submitted – International Conference on Innovations in Electronics, Communication & Computing Techniques (ICIECT), Paper ID 397

- Evaluated XAI methods (Grad-CAM, Grad-CAM++, LIME, KernelSHAP) on pneumonia classification using a fine-tuned EfficientNet-B0 model trained on 5,200+ images.
- Optimized model performance to 90.87% accuracy and 0.9693 AUC-ROC using staged fine-tuning, weighted loss, and Test-Time Augmentation (TTA).
- Quantified faithfulness and sparsity of XAI outputs; identified KernelSHAP as the most sparse (0.878) and used error analysis to mitigate spurious feature correlations.

PROJECT

AetherDrop — Secure File Sharing Platform

github.com/NipunXD/aetherdrop

- Built and deployed a full-stack file-sharing system supporting multi-file uploads, expiring links, and download limits.
- Designed backend APIs using Node.js and Express to enforce access control, expiry validation, and download tracking.
- Integrated AWS S3 for secure file storage, ensuring no direct public access to files.
- Implemented on-the-fly ZIP streaming for multi-file downloads using Archiver.
- Containerized the entire system using Docker and deployed on AWS EC2 with Nginx reverse proxy.
- Configured routing to handle frontend-backend communication via /api endpoints, eliminating CORS issues.
- Exposed the application securely using Cloudflare Tunnel with HTTPS support.

Patient Risk Stratification - Clinical AI Analyzer

github.com/NipunXD/Patient-Risk-Stratification

- Built an end-to-end clinical risk stratification system to predict 30-day hospital readmissions using unstructured clinical notes.
- Fine-tuned ClinicalBERT on the MIMIC-III dataset to interpret medical jargon and contextual negation.
- Architected a decoupled microservice system with a Python Flask inference API and a Next.js clinician dashboard.
- Achieved 90% accuracy on qualitative test sets and proposed hybrid models for Social Determinants of Health (SDoH).

Maitri-AI - Multimodal Offline Well-Being Assistant for Astronauts

github.com/NipunXD/Maitri-AI

- Developed an offline-first AI assistant to support psychological well-being in constrained environments.
- Integrated LLaMA via local inference (Ollama) with sentiment-aware prompting for conversational support.
- Implemented facial emotion detection using DeepFace and OpenCV for real-time analysis.
- Built a React/Next.js interface with structured assessments and session tracking.

TruthGuard - NLP-Based Fake News Detection Engine

github.com/NipunXD/TruthGuard

- Developed an AI-powered system to classify news articles as real or fake with confidence scores.
- Trained and evaluated BERT models on the WELFake dataset (72,134 articles) to reduce classification bias.
- Implemented explainability features to improve transparency and user trust.
- Designed as a platform-agnostic engine for easy integration with social platforms and news aggregators.

EcoRate — Lifestyle Sustainability Assessment

github.com/NipunXD/EcoRate

- Designed a machine learning pipeline to predict numerical sustainability ratings based on individual lifestyle data.
- Performed feature engineering and model selection across regression and ensemble approaches.
- Achieved $R^2 = 0.71$ and $MSE = 0.73$, demonstrating strong predictive performance.
- Built a user-facing dashboard to visualize scores and deliver personalized recommendations.

Heritage Vault - Secure Provenance Tracking (SIH 2024)

github.com/NipunXD/HeritageVault

- Proposed a blockchain-based platform to securely track the provenance of cultural artifacts.
- Designed an immutable ledger to record ownership history, addressing fraud and illicit trade.
- Integrated digital fingerprinting and smart contracts for automated ownership transfers using Next.js, TypeScript, and Solidity.

SKILLS

Languages: Python, C++, TypeScript, SQL

Frontend: React, Next.js, Tailwind CSS

Backend: Node.js, Express.js, Flask

Cloud & Deployment: AWS S3, EC2, Docker, Nginx

ML: NLP (BERT), Regression, Classification

Tools: Git, VS Code, XCode

CERTIFICATIONS

- Data Structures — University of California San Diego (Coursera)
- Object-Oriented Data Structures in C++ — University of Illinois (Coursera)
- Databases and SQL for Data Science with Python — IBM (Coursera)
- Machine Learning: Regression — University of Washington (Coursera)
- Concours Romain Rolland French Language Competition — AIR 117, Level B1